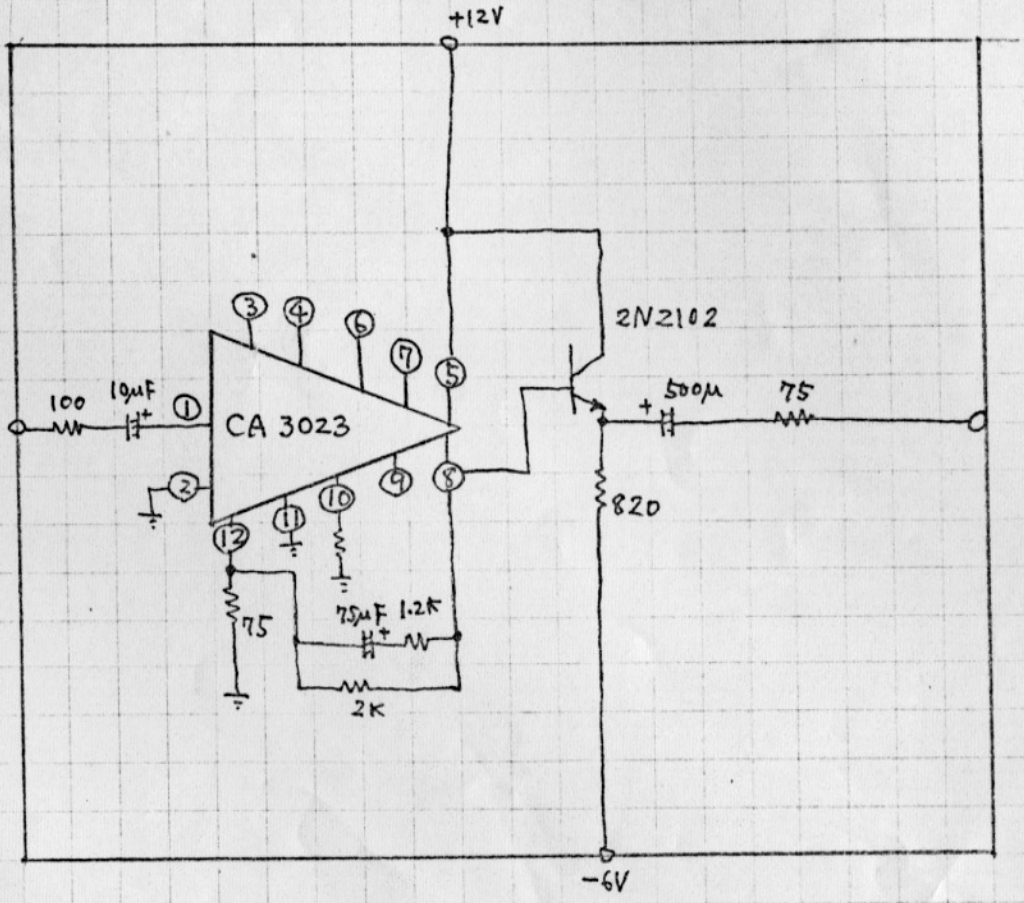
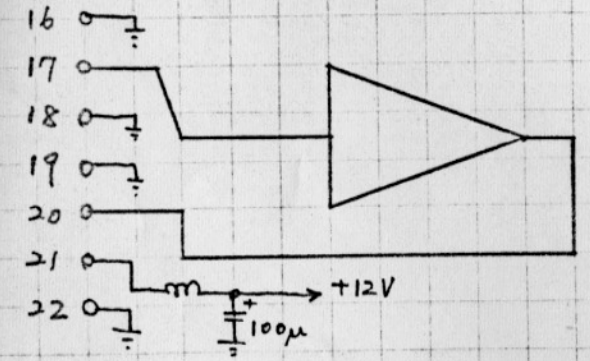
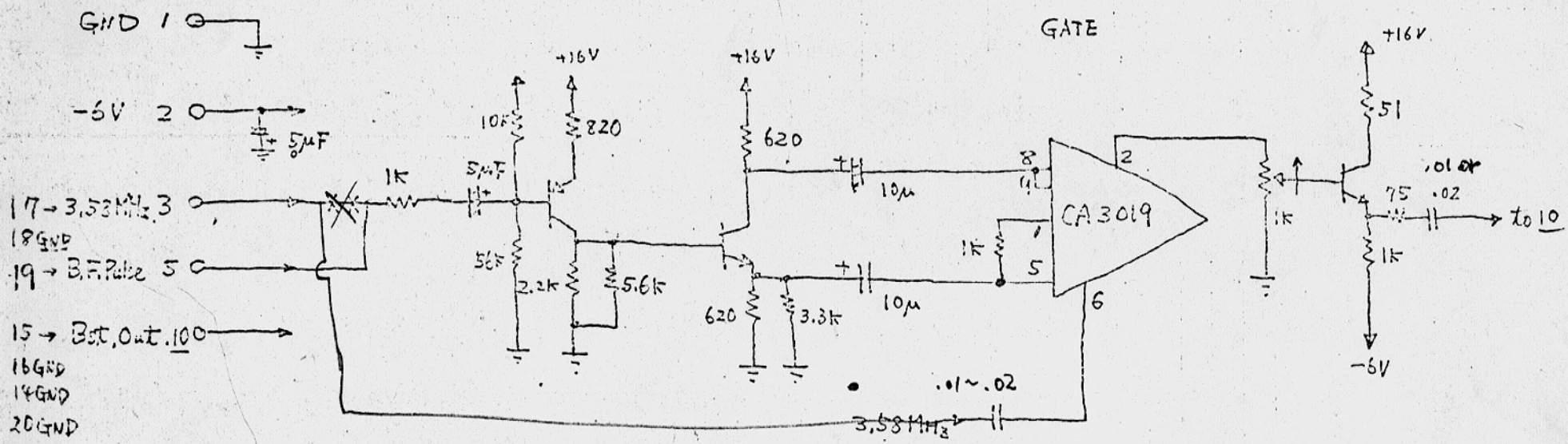
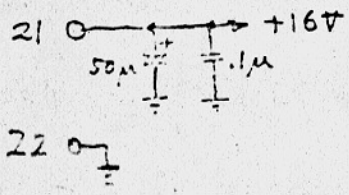

1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
-6V
100μ


+12V
CA 3023
2N2102
100
10μF
500μ
75
820
75
75μF
1.2K
2K
-6V

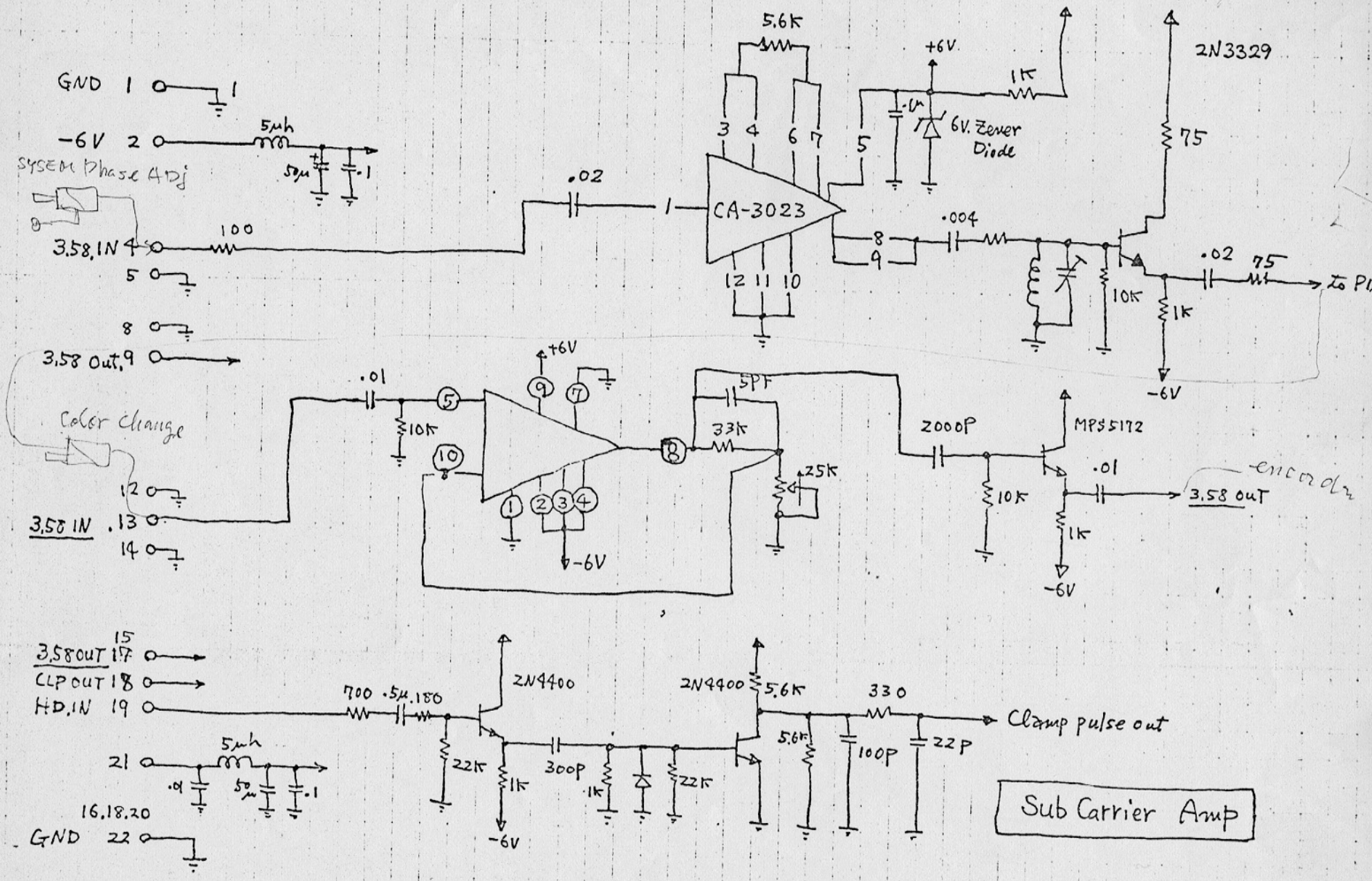

+12V
100μ

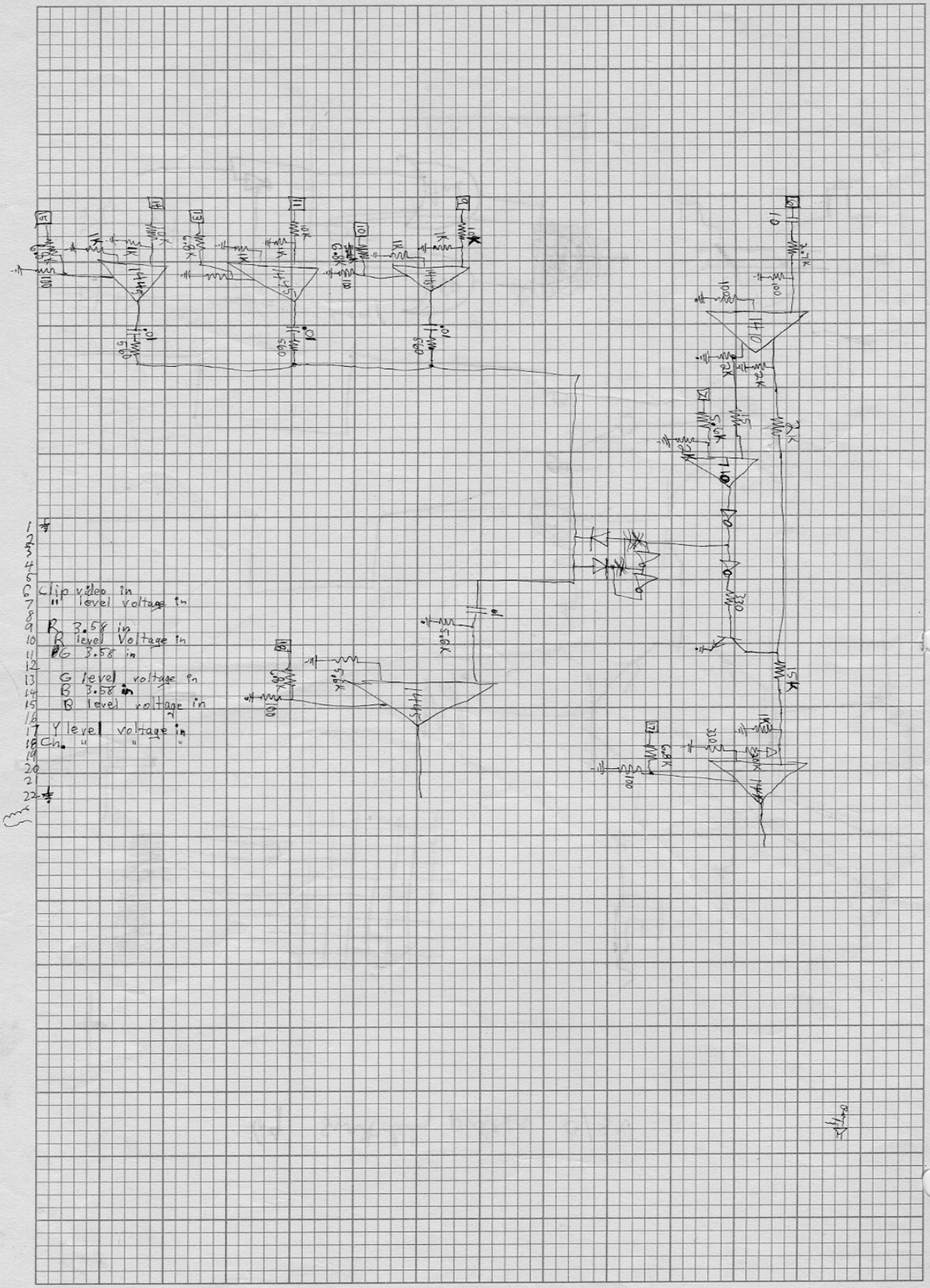
Amp.

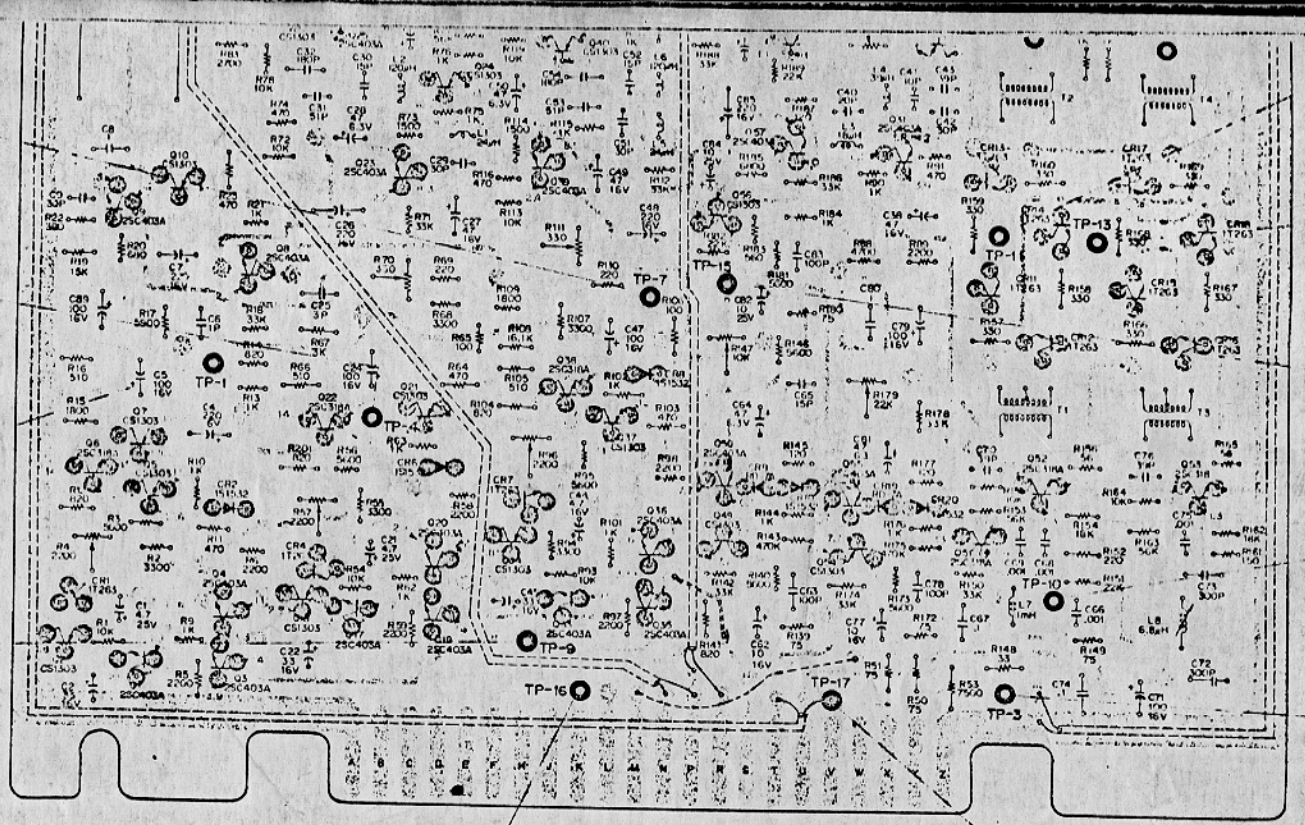


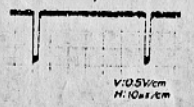


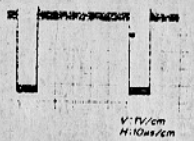
Burst Generator


GND 1
-6V 2
5μh
SYSEM Phase Adj
3.58 IN 4
5
8
3.58 Out 9
Color Change
12
3.58 IN 13
14
100
.02
CA-3023
5.6K
+6V
1K
6V. Zener Diode
2N3329
75
.004
10K
.02
75
To PL
-6V
.01
10K
+6V
5PF
33K
25K
-6V
2000P
MPS5172
.01
3.58 OUT
encoder
15
3.58OUT 17
CLP OUT 18
HD. IN 19
21
16.18.20
GND 22
700
.5μ.180
2N4400
22K
1K
-6V
300P
1K
22K
2N4400
5.6K
330
5.6K
100P
22P
Clamp pulse out
Sub Carrier Amp


Clip video in
" level voltage in
B 3.58 in
R level voltage in
G 3.58 in
G level voltage in
B 3.58 in
R level voltage in
Y level voltage in
Ch. " " "



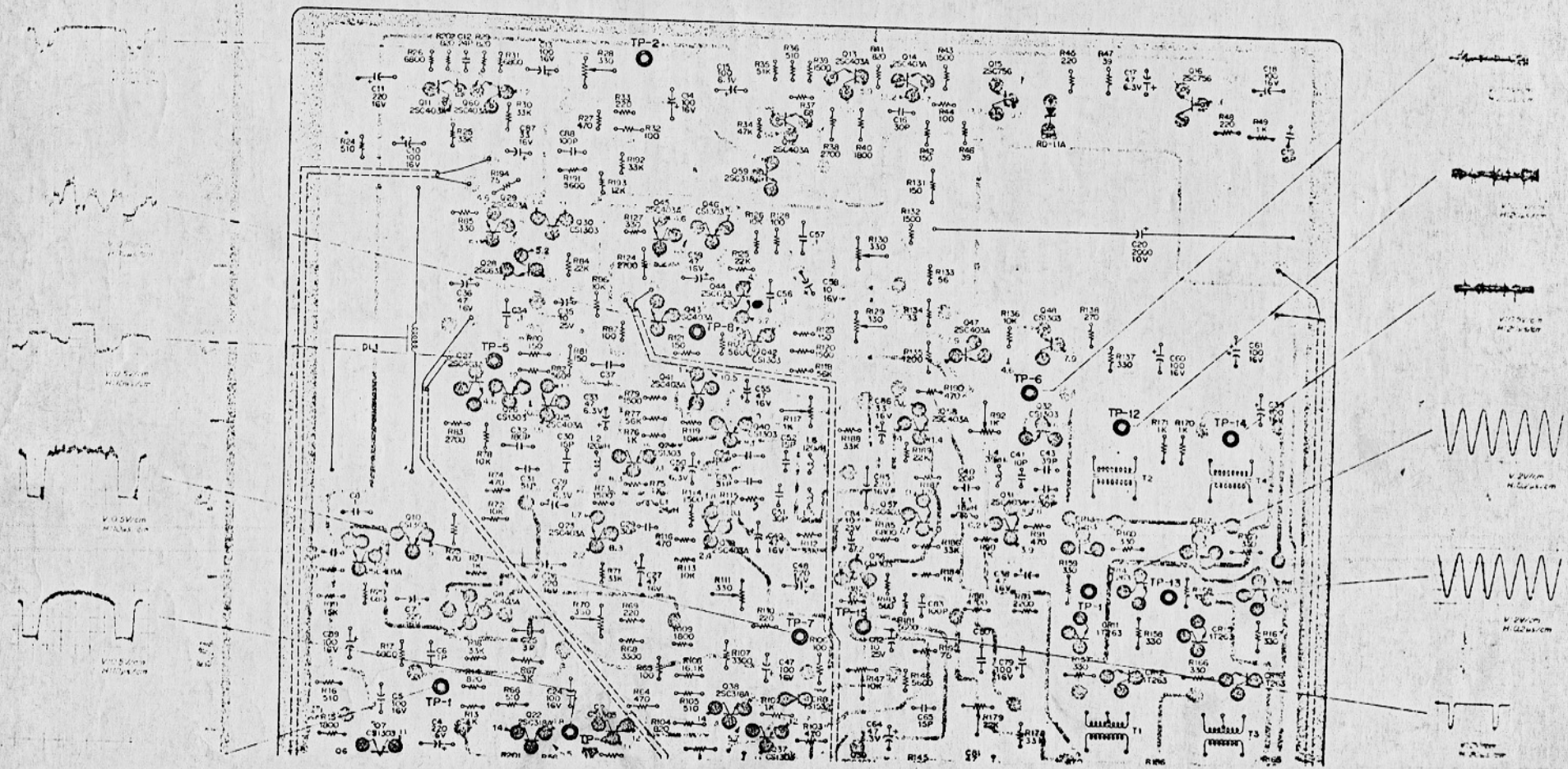


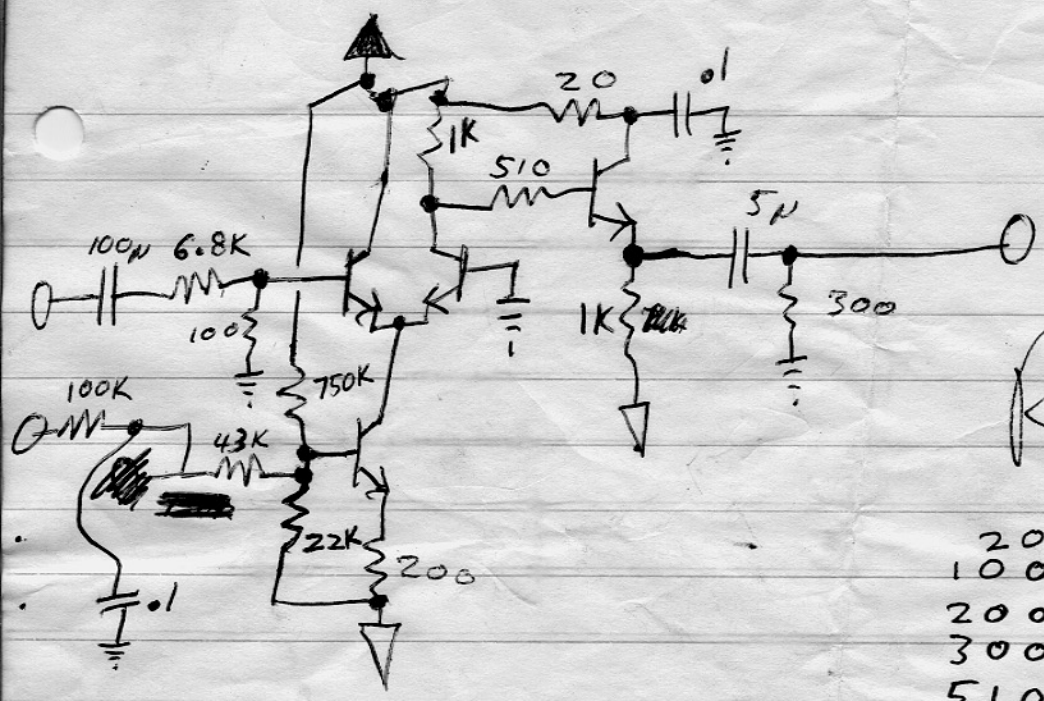


Note:
Printed pattern of component side is shown in gray.

. ENCODER

—conducter side—





R

20 — 1
100 — 1
200 — 1
300 — 1
510 — 1

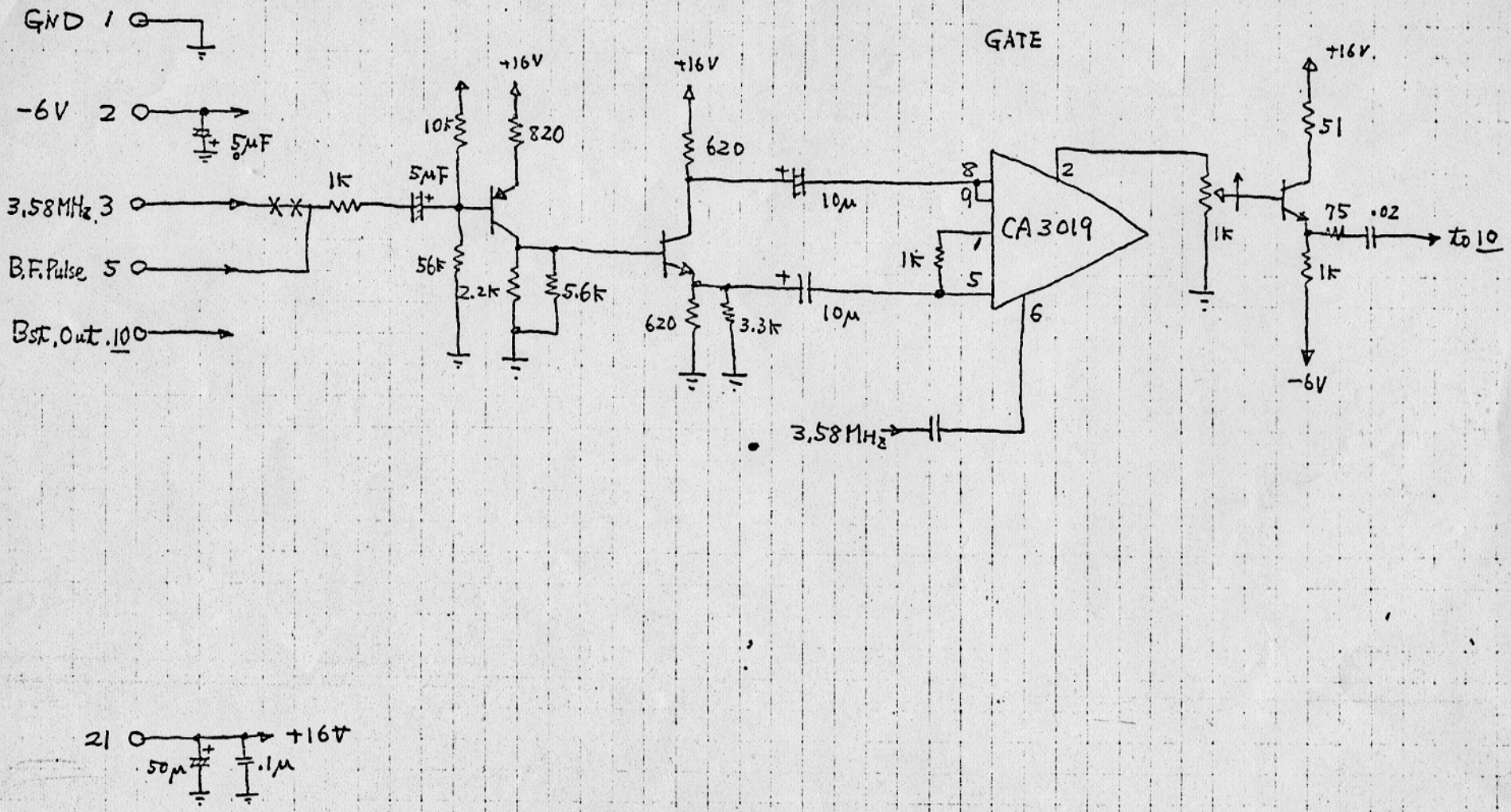
1K — 2

6.8K — 1
22K — 1
43K — 1
100K — 1
750K — 1

C

.1 — 2
5 — 1
100 — 1

PN918 - 3

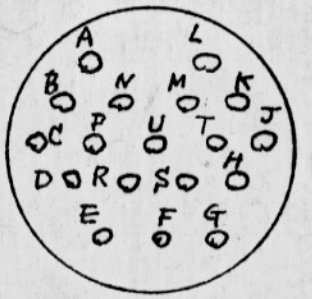


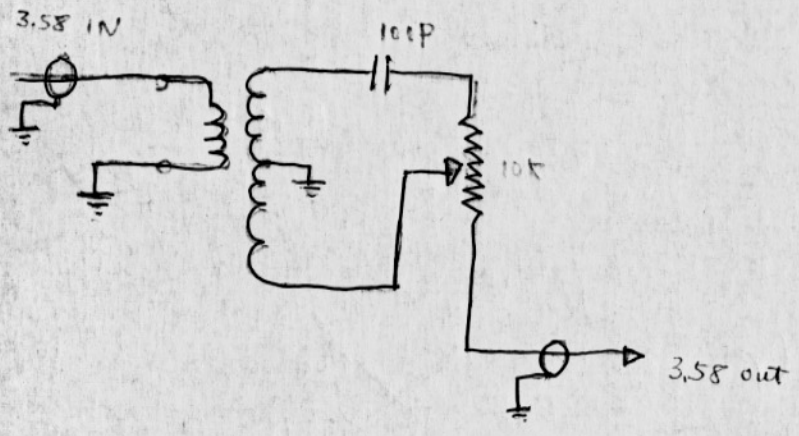
Burst Generator

A + 12V
B − 6V
P GND
C Rout

E Coax SEL VID
H F out
L T out

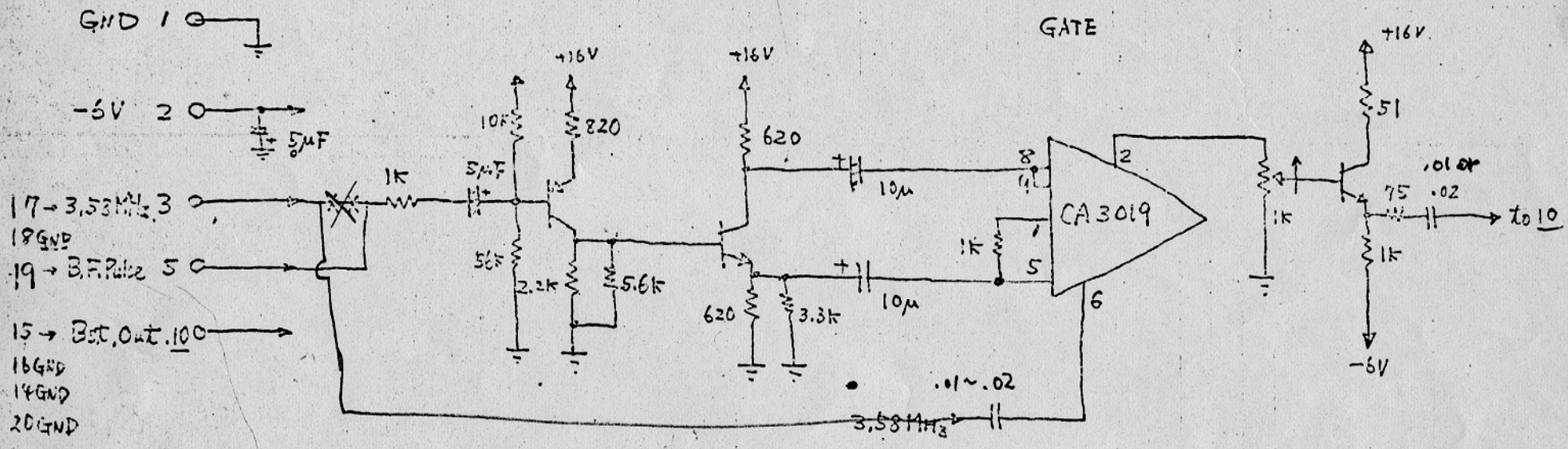
F GND
K "

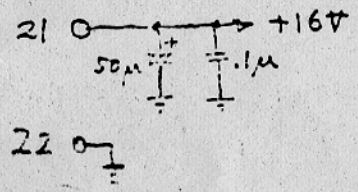




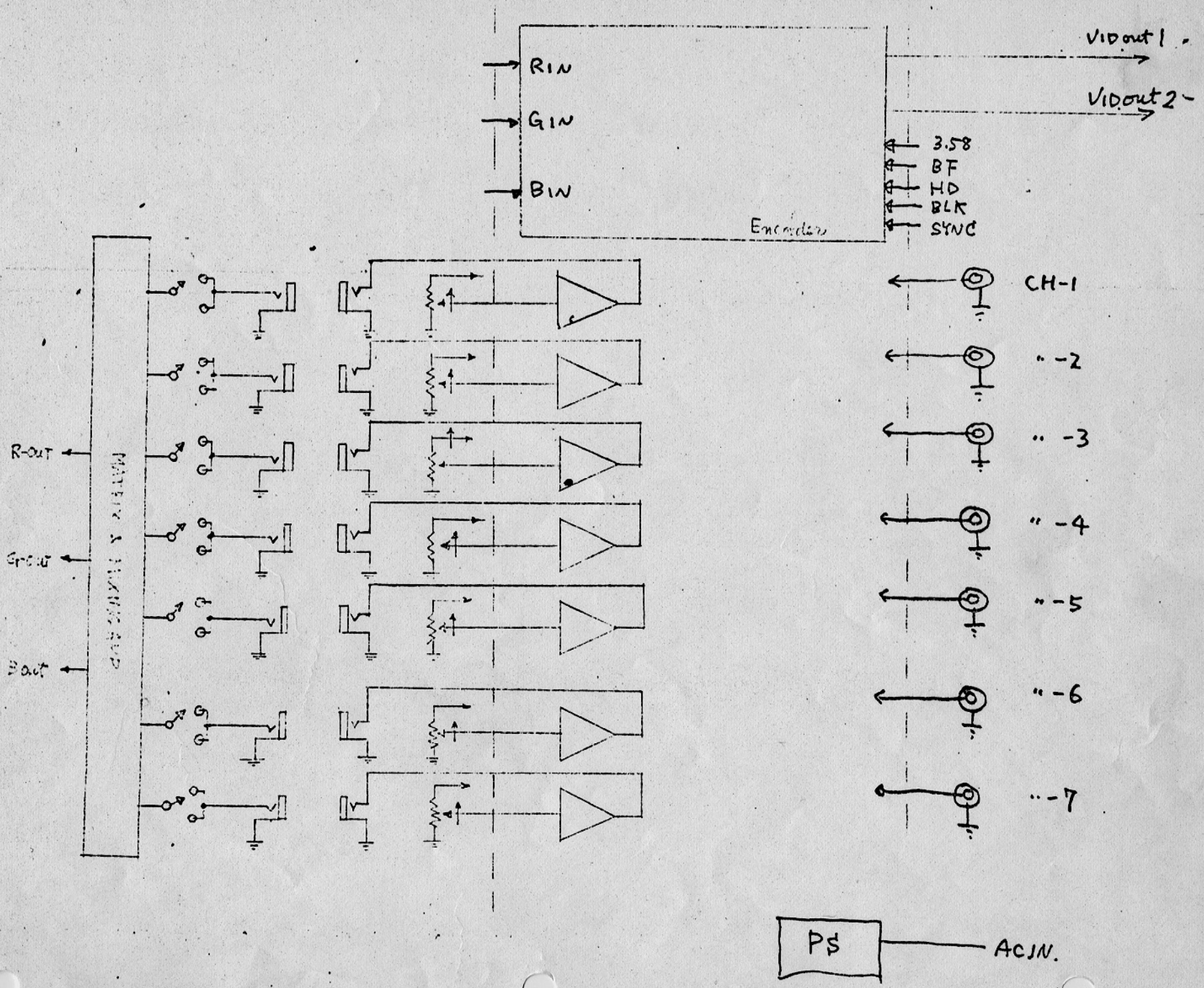
Phase CTL.

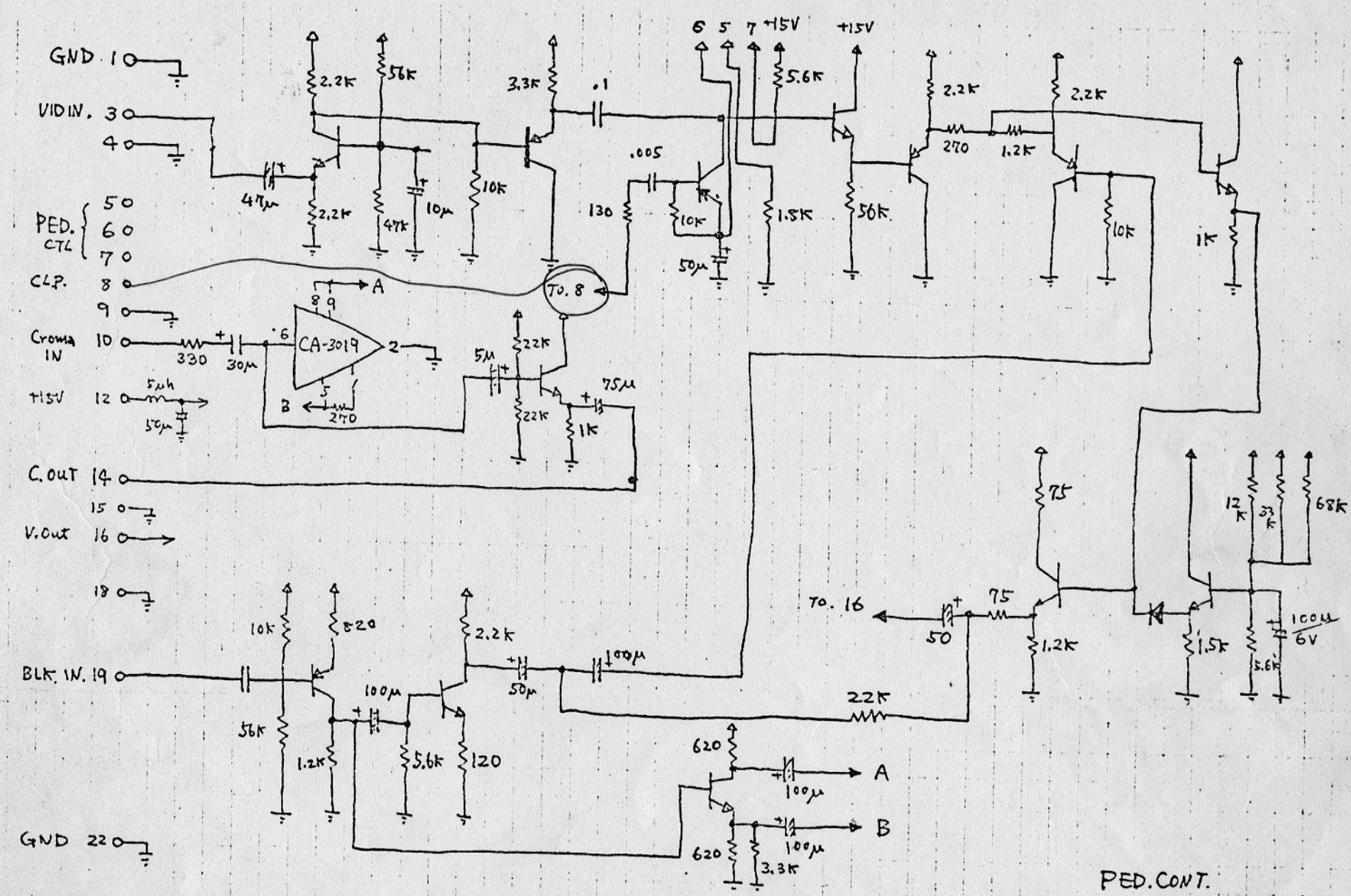
CONNETOR Diagram

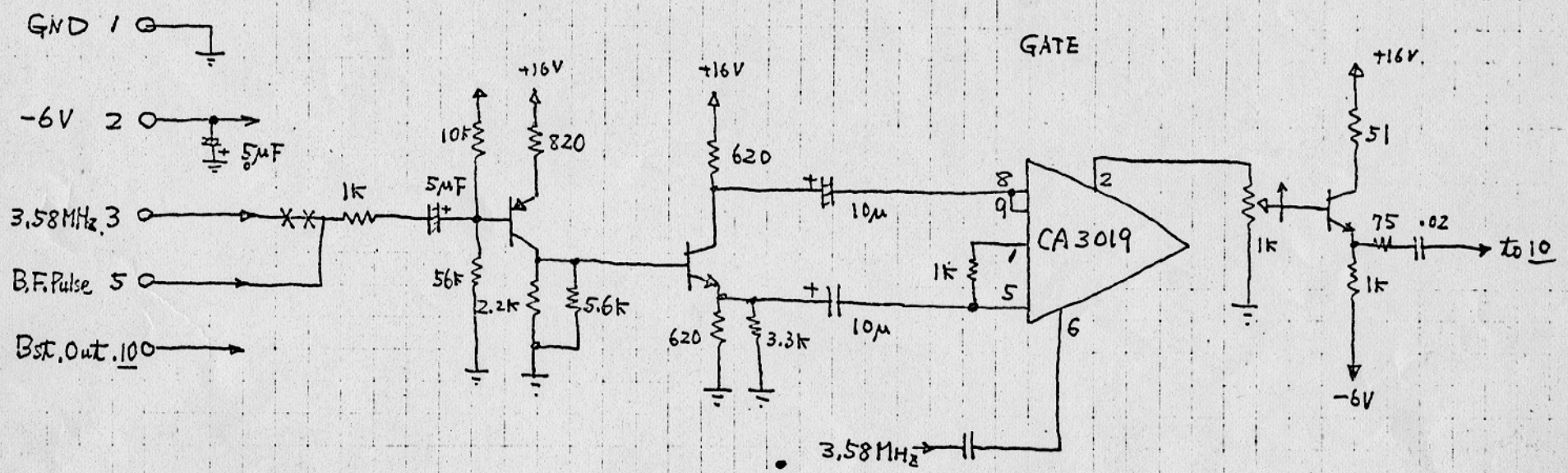


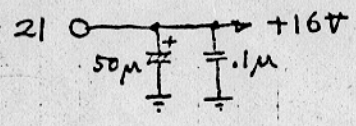


Burst Generator

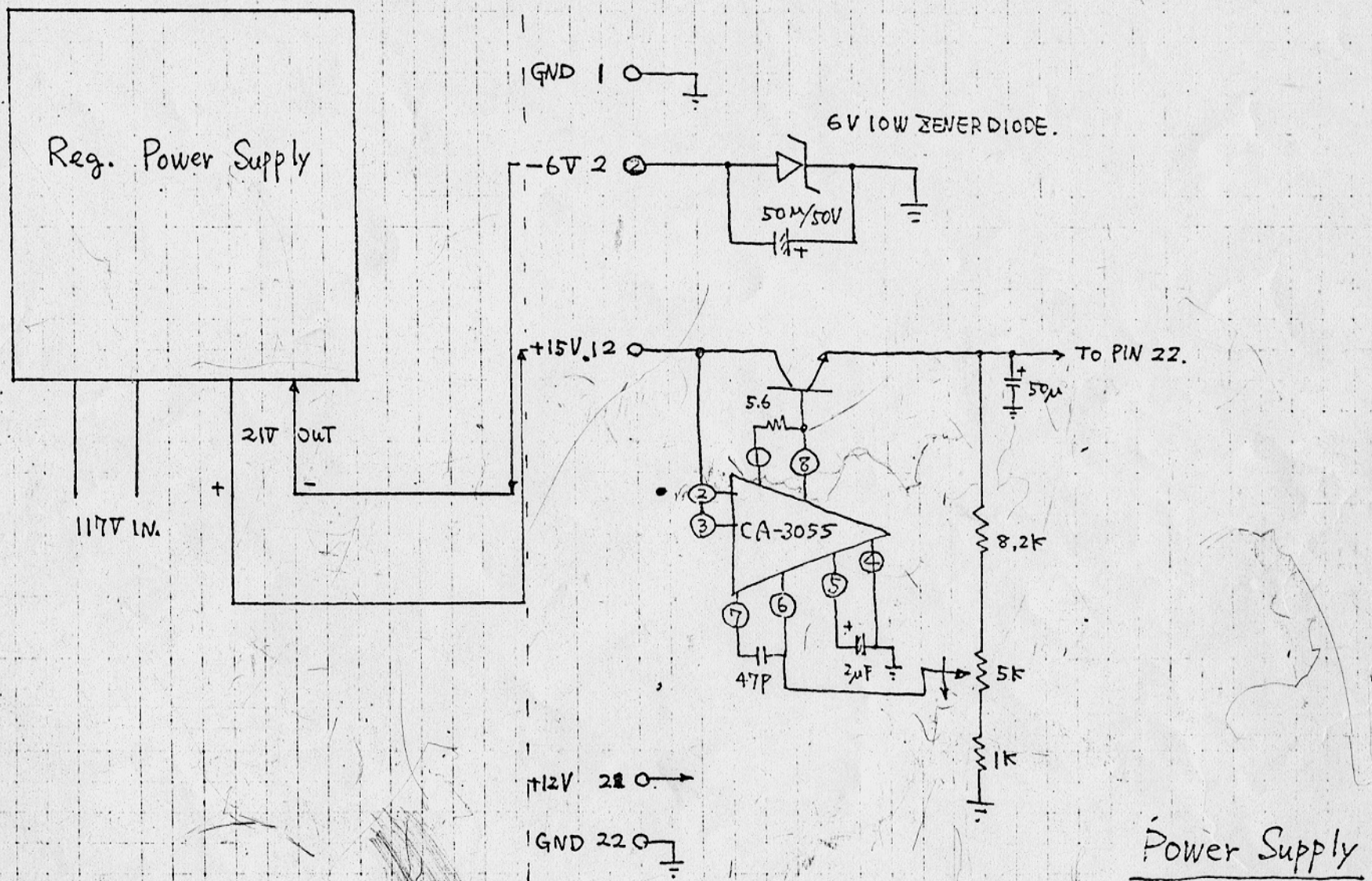

RIN
GIN
BIN
Encoder
VIDout 1
VIDout 2
3.58
BF
HD
BLK
SYNC
CH-1
" -2
" -3
" -4
" -5
" -6
" -7
R-OUT
PS
AC IN.


GND 1
VID IN. 3
4
PED. CTL 5 6 7
CLP. 8
9
Croma IN 10
+15V 12
C. OUT 14
15
V. Out 16
18
BLK. IN. 19
GND 22
CA-3019
To. 8
To. 16
A
B
+15V
2.2K
56K
3.3K
.1
.005
130
10K
47µ
47K
10µ
1.5K
5.6K
270
1.2K
1K
330
30µ
5µh
50µ
22K
5M
75µ
75
12K
33K
68K
100µ 6V
50
100µ
620
3.3K
120
PED. CONT.

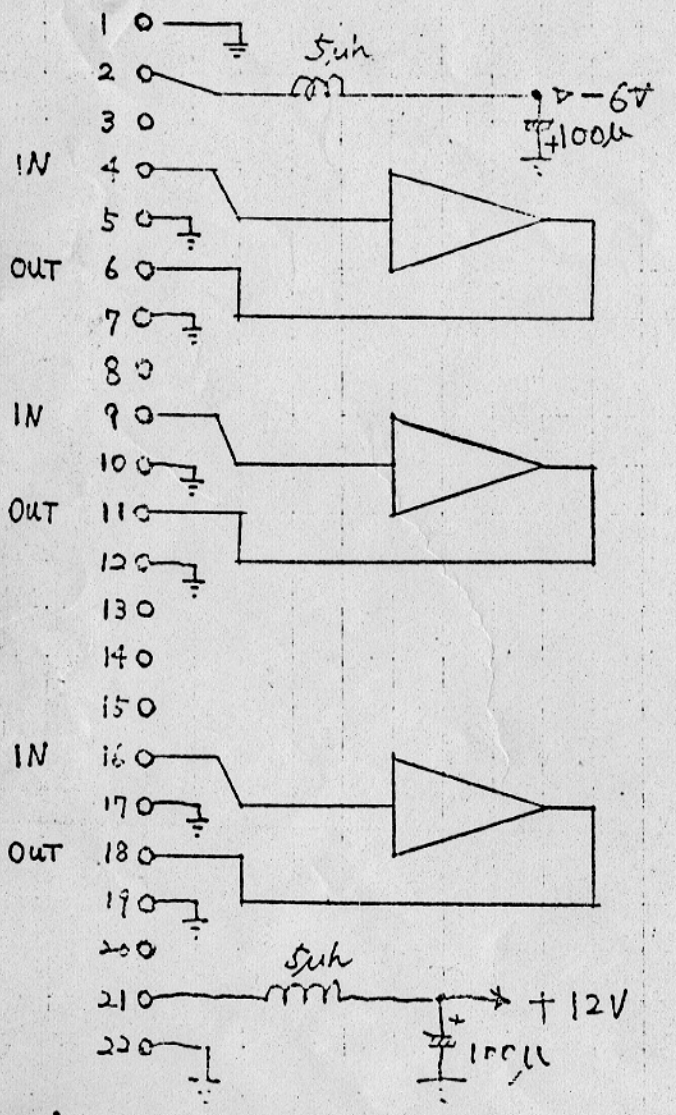


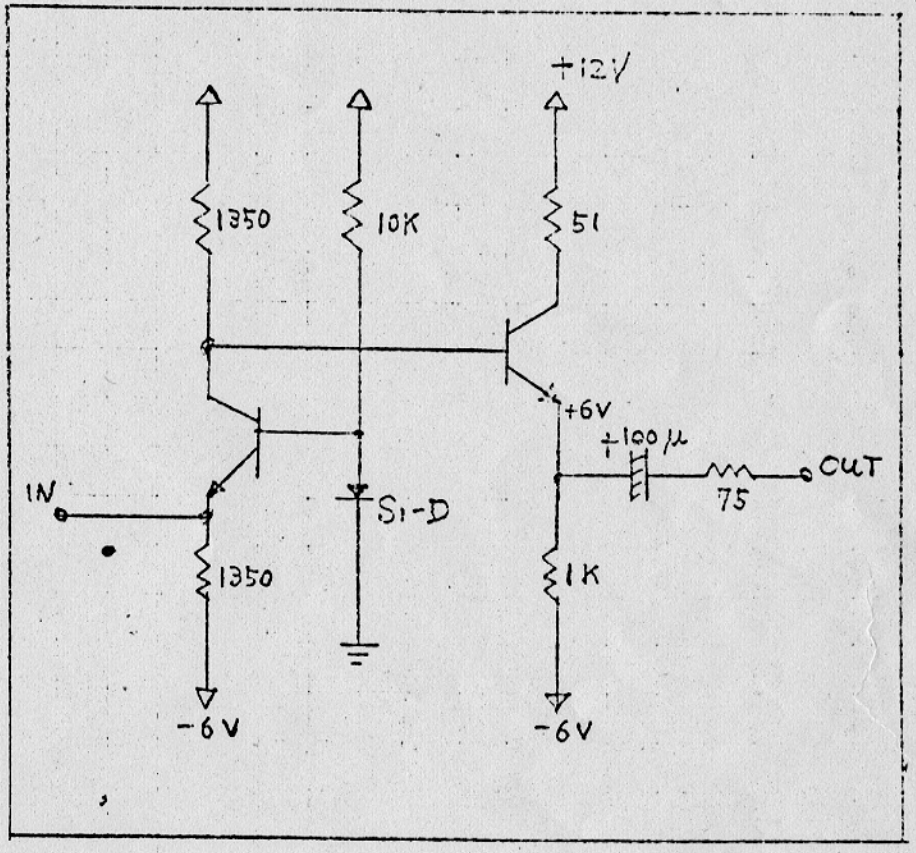


Burst Generator

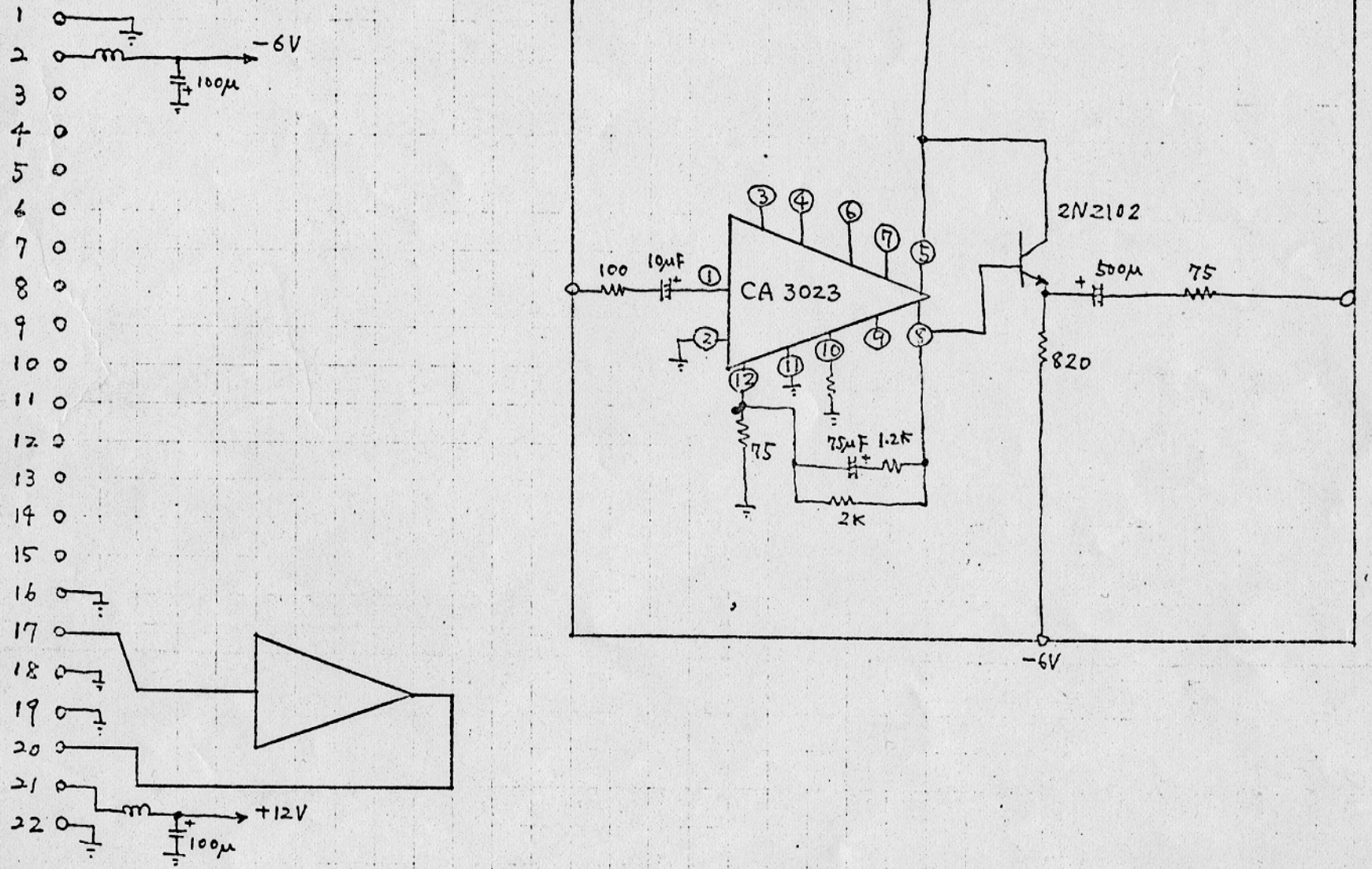


Power Supply

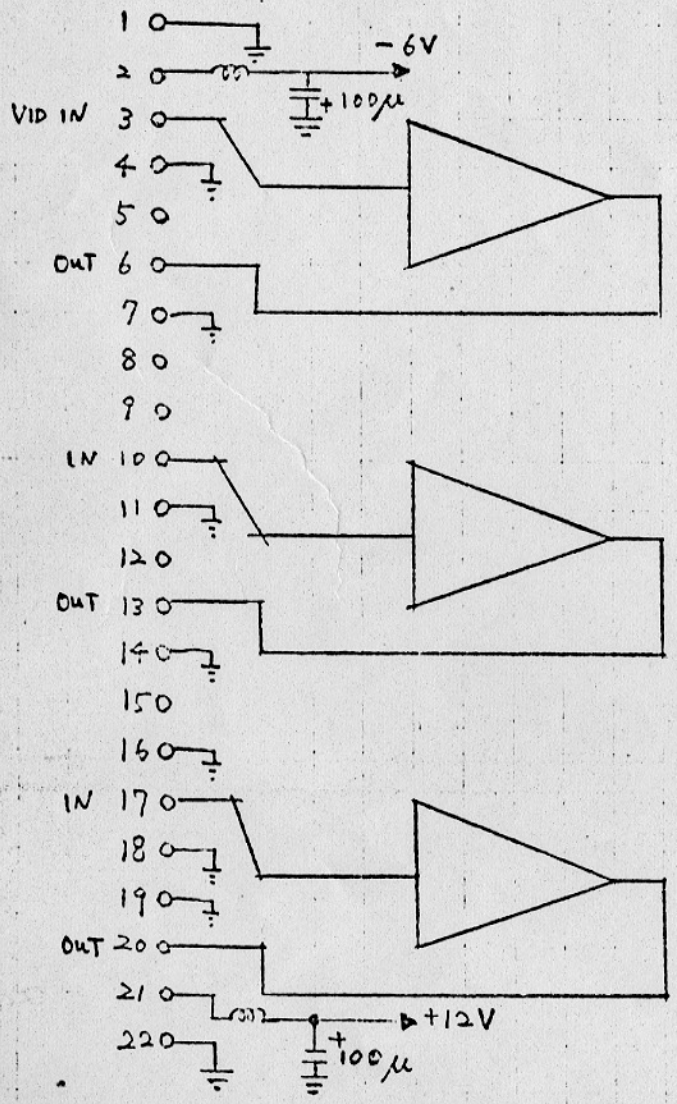

1
2
3
IN 4
5
OUT 6
7
8
IN 9
10
OUT 11
12
13
14
15
IN 16
17
OUT 18
19
20
21
22
5uh
-6V
+100u
5uh
+12V
100u

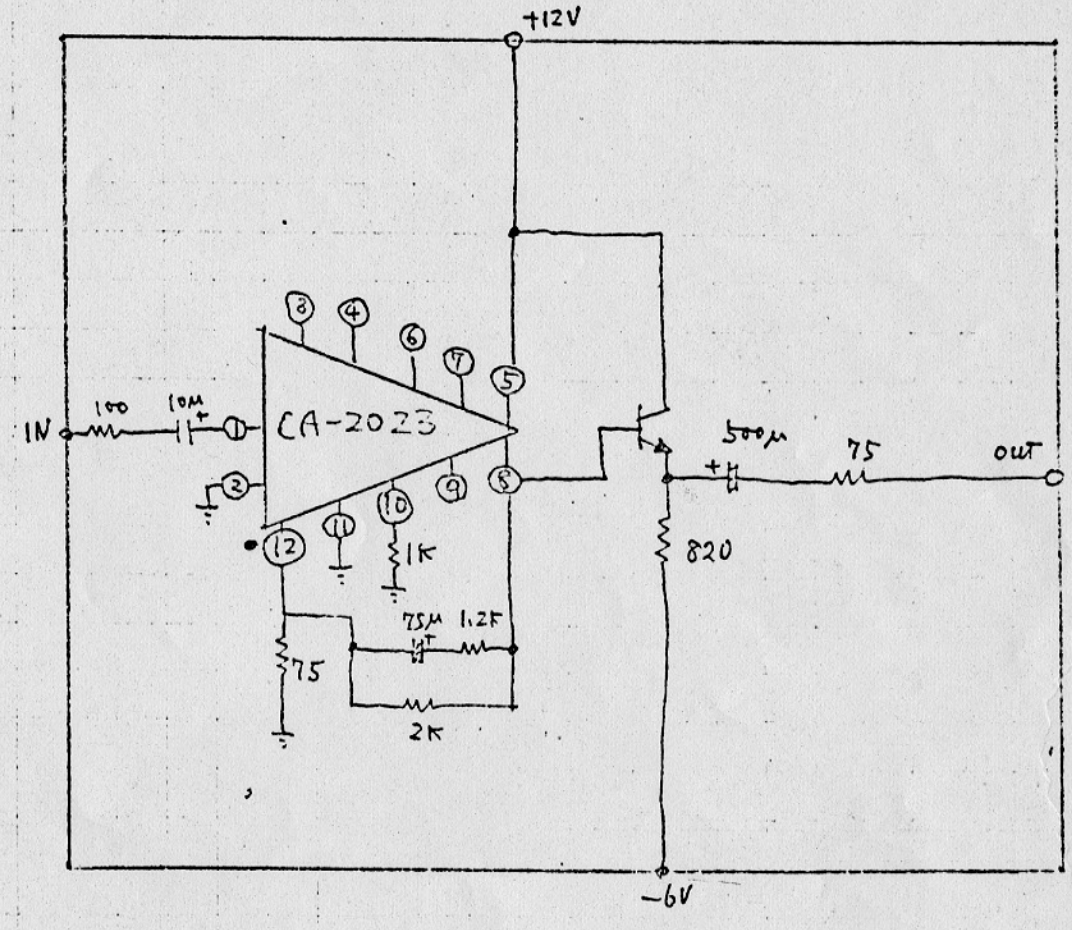

+12V
1350
10K
51
+6V
+100μ
IN
S1-D
75
OUT
1350
1K
-6V
-6V

Mixing Amp

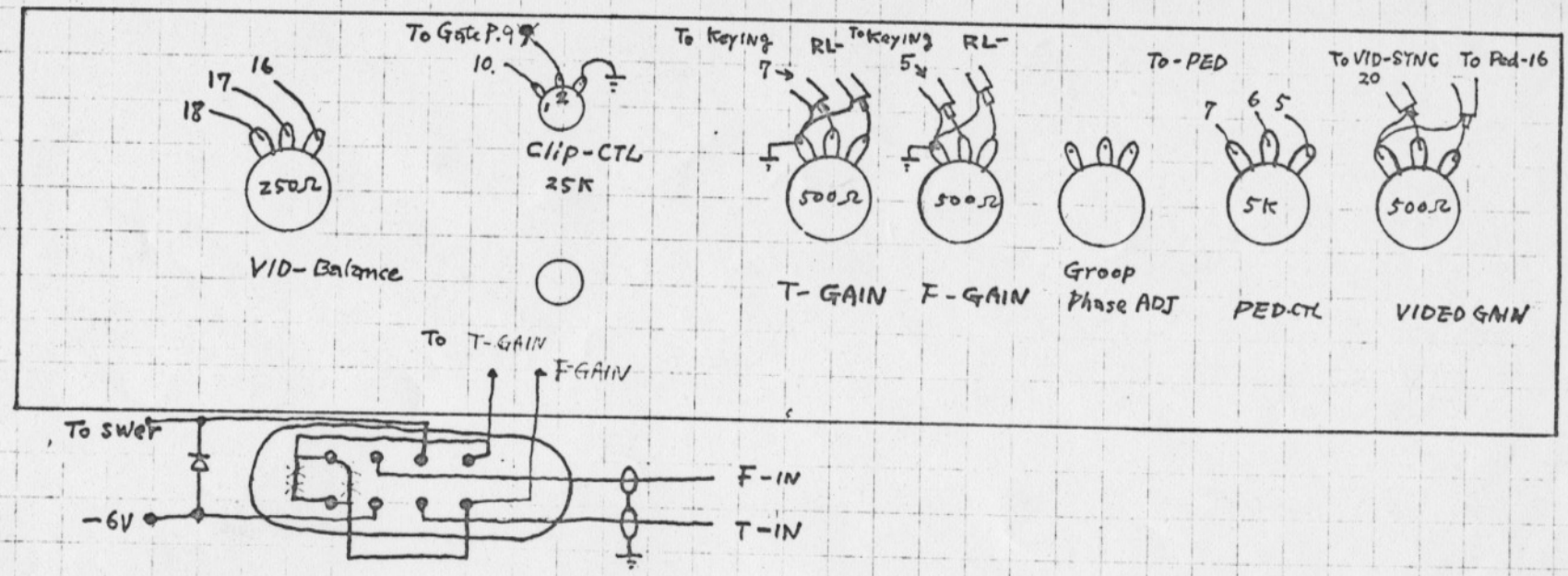

1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
-6V
100μ
+12V
100μ
+12V
CA 3023
100
10μF
2N2102
500μ
75
820
75
75μF
1.2K
2K
-6V

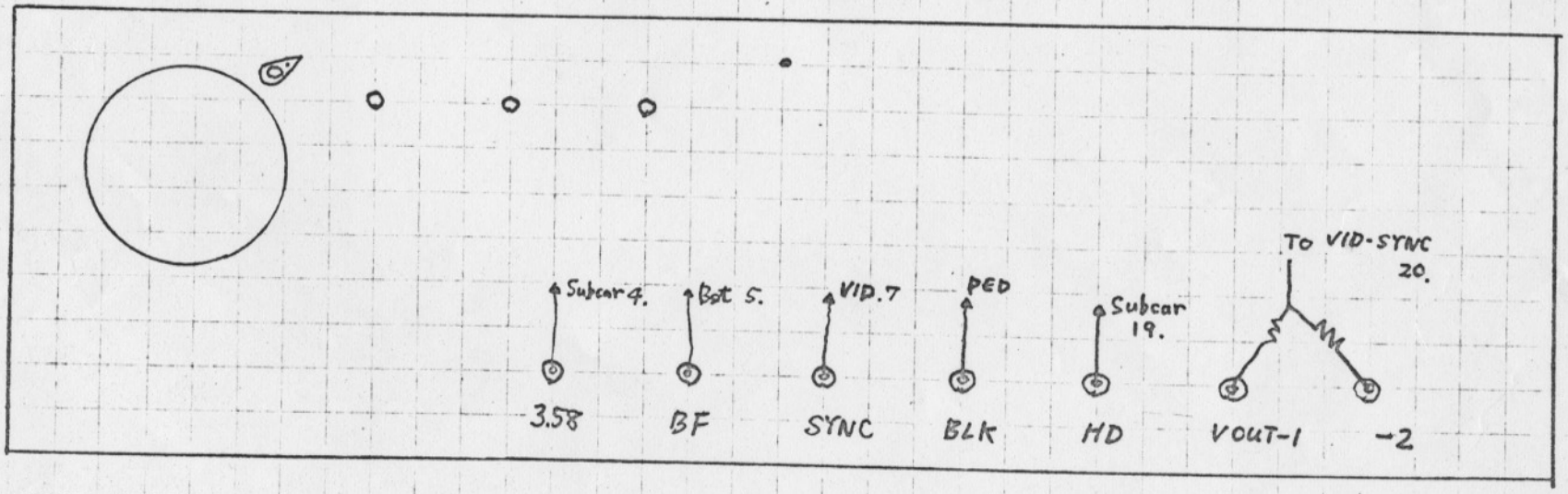
Amp.

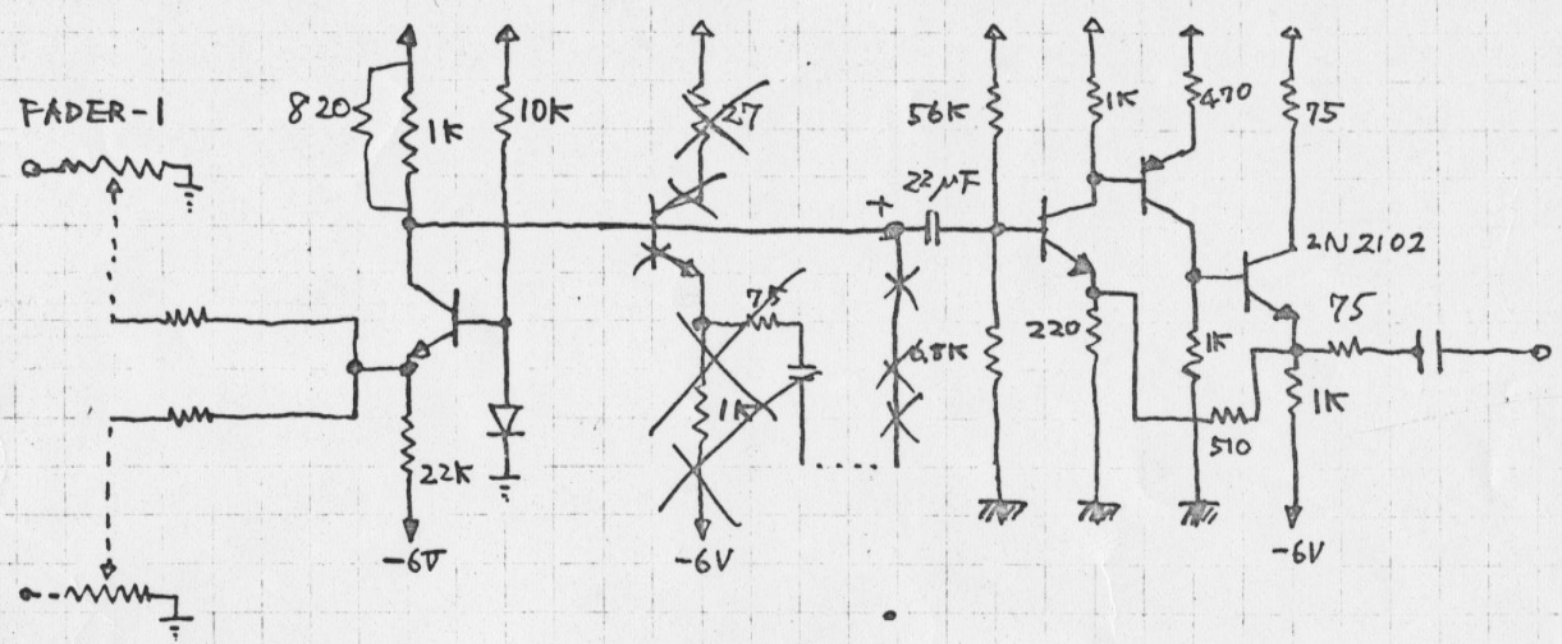

1
2
VID IN 3
4
5
OUT 6
7
8
9
IN 10
11
12
OUT 13
14
15
16
IN 17
18
19
OUT 20
21
22
-6V
+100μ
+12V
+100μ


+12V
IN
100
10M
CA-2023
500μ
75
OUT
820
1K
75M
1.2K
75
2K
-6V

Amp.


To Gate P.9
10.
Clip-CTL
25K
To Keying
7
RL-
To Keying
5
RL-
To-PED
To VID-SYNC
20
To Ped-16
18
17
16
250Ω
VID-Balance
500Ω
T-GAIN
500Ω
F-GAIN
Group
Phase ADJ
7
6
5
5K
PED-CTL
500Ω
VIDEO GAIN
To T-GAIN
F-GAIN
To Swep
-6V
F-IN
T-IN


Subcar 4.
Bst 5.
VID.7
PED
Subcar
19.
To VID-SYNC
20.
3.58
BF
SYNC
BLK
HD
VOUT-1
-2

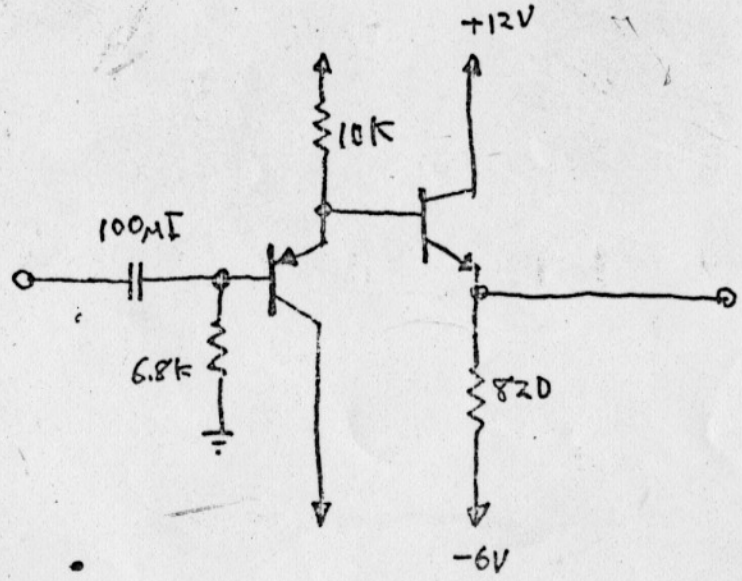


FADER - 2

PNP ... 2N4402 or Equivalent

NPN ... 2N4400 " "

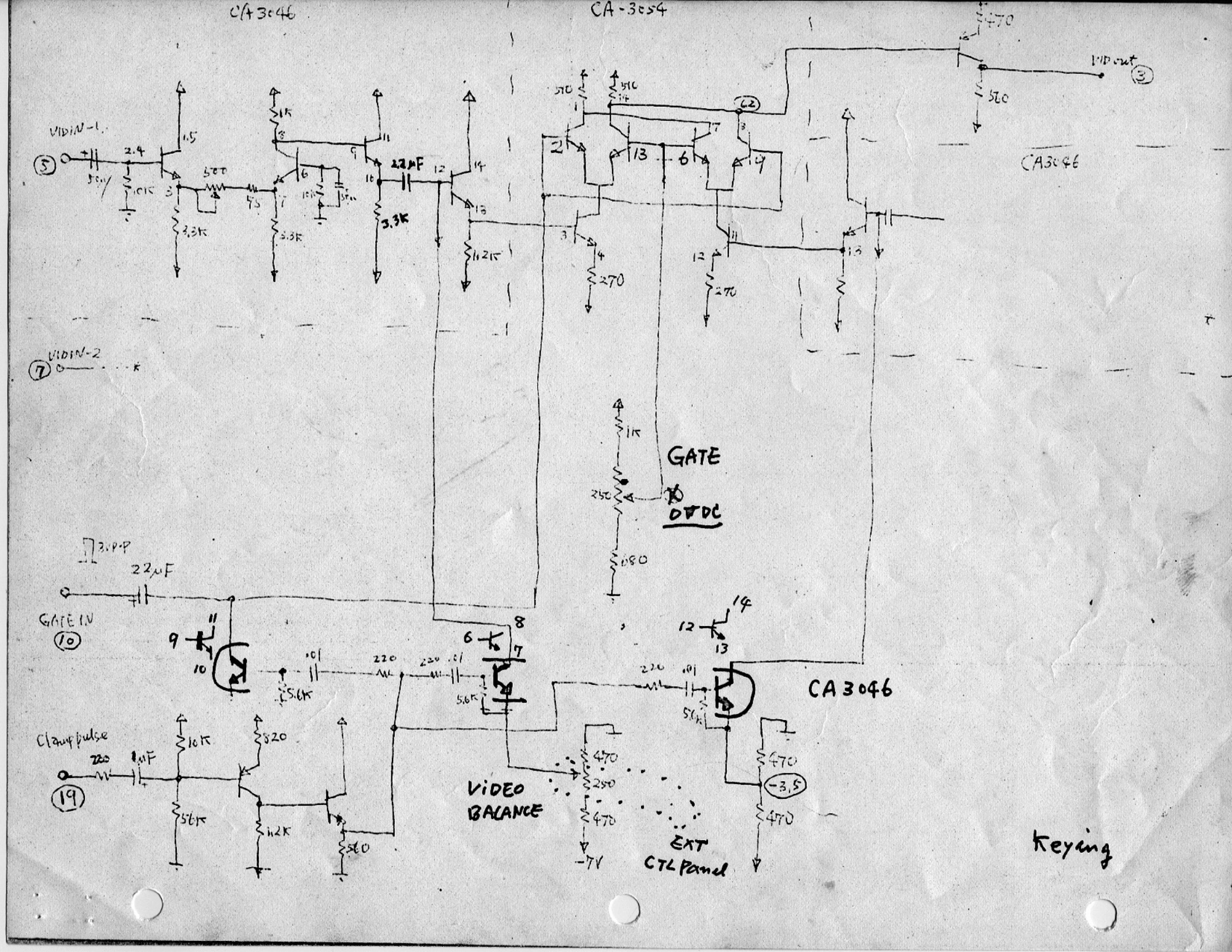
MIXER - FADER OUTPUT AMP



SAME AS Above

SAME AS Above

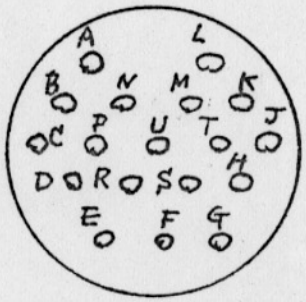
SWITCHER INPUT
EMITTER
Follower


CA3046
CA-3054
VIDIN-1
VIDIN-2
VID out
CA3046
GATE
0 VDC
GATE IN
22μF
3 P-P
Clamp pulse
VIDEO BALANCE
EXT CTL Panel
-7V
-3.5
CA 3046
Keying

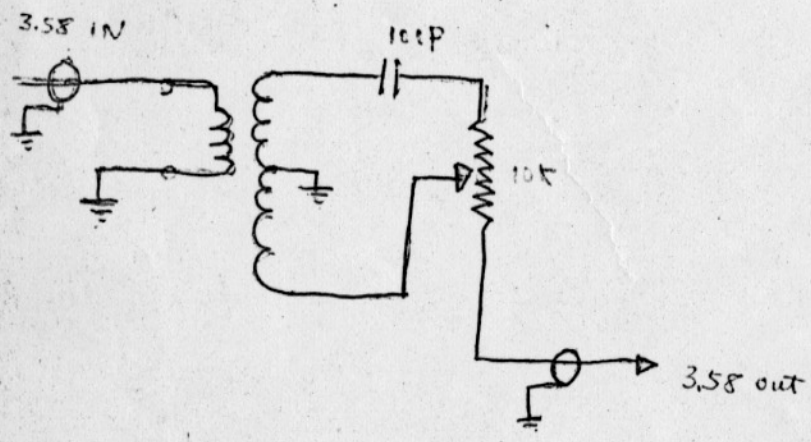
A + 12V
B − 6V
P GND
C Rout

E Coax SEL VID
H F out
L T out

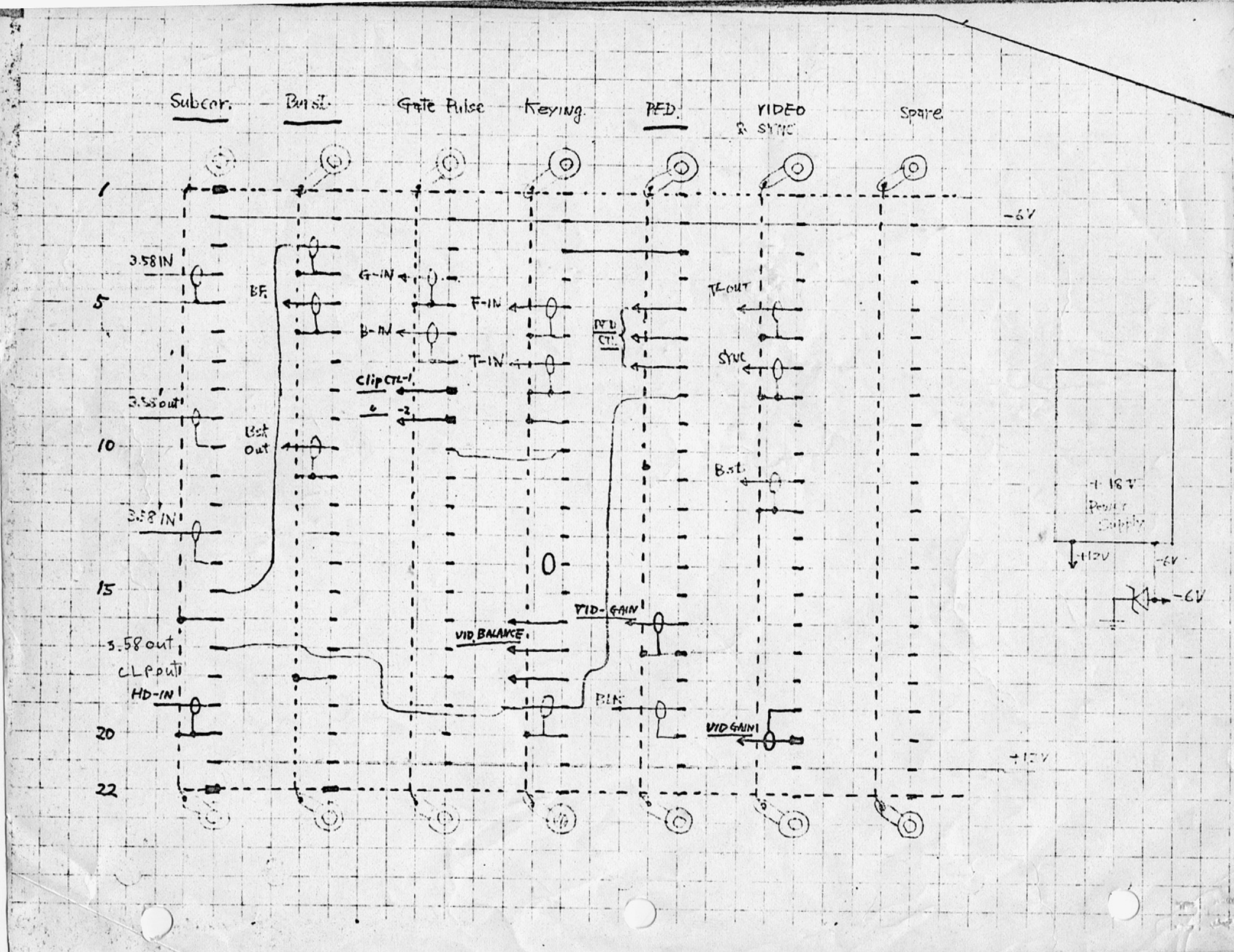
F GND
K "

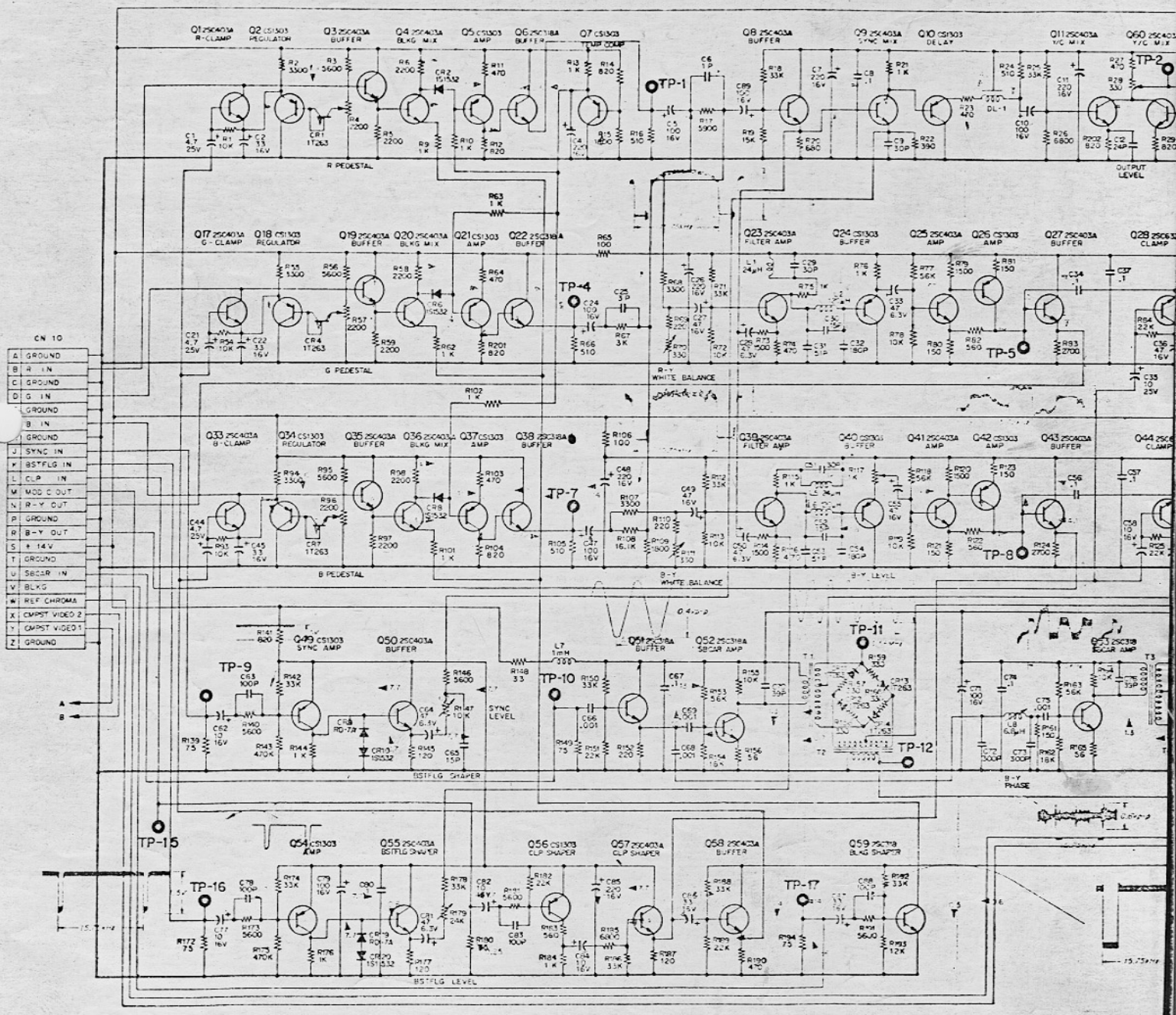


CONNECTOR Diagram

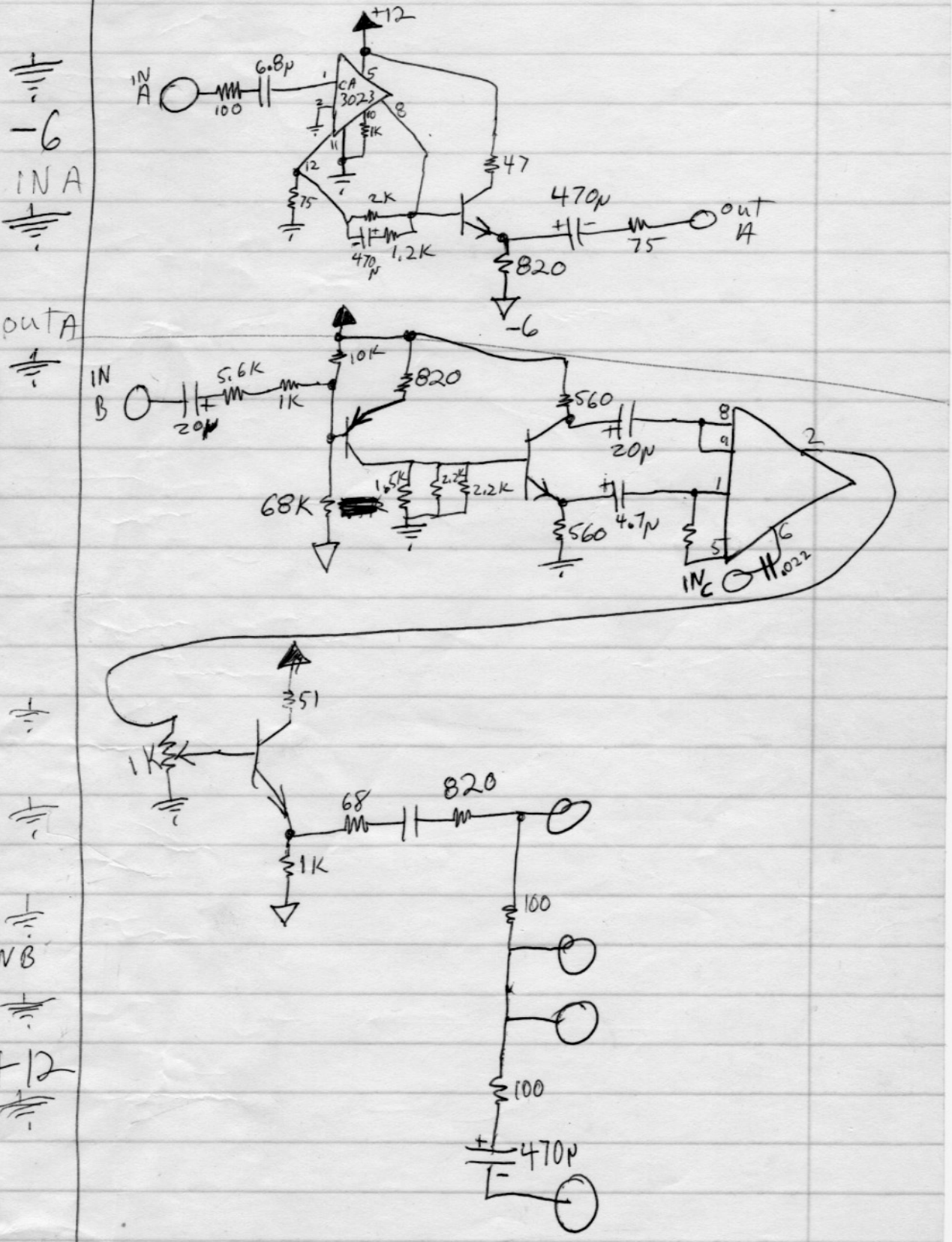


Phase CTL


Subcar.
Burst
Gate Pulse
Keying
PED.
VIDEO & SYNC
Spare
1
5
10
15
20
22
-6V
+12V
3.58 IN
BF
3.58 out
Burst Out
3.58 IN
3.58 out
CLP out
HD-IN
G-IN
F-IN
B-IN
T-IN
Clip CTL-1
-2
VID BALANCE
BLK
TL-OUT
PED CTL
SYNC
Burst
VID GAIN
VID-GAIN
0
+18V Power Supply
+12V
-6V
-6V

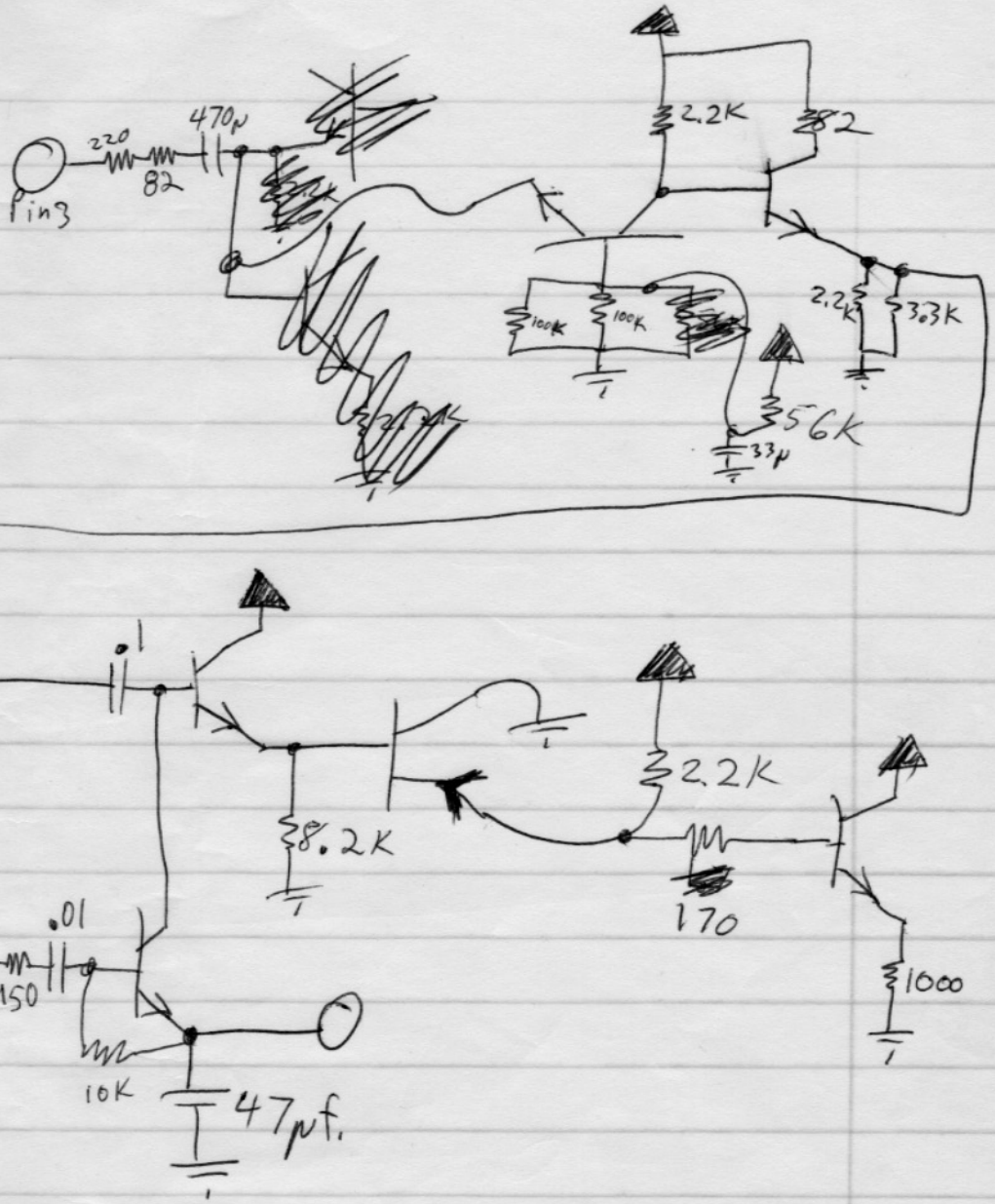

CN 10
A GROUND
B R IN
C GROUND
D G IN
E GROUND
F B IN
G GROUND
J SYNC IN
K BSTFLG IN
L CLP IN
M MOD C OUT
N R-Y OUT
P GROUND
R B-Y OUT
S + 14V
T GROUND
U SBCAR IN
V BLKG
W REF CHROMA
X CMPST VIDEO 2
Y CMPST VIDEO 1
Z GROUND
TP-1
TP-2
TP-4
TP-5
TP-7
TP-8
TP-9
TP-10
TP-11
TP-12
TP-15
TP-16
TP-17
R PEDESTAL
G PEDESTAL
B PEDESTAL
R-Y WHITE BALANCE
B-Y WHITE BALANCE
B-Y LEVEL
SYNC LEVEL
BSTFLG SHAPER
BSTFLG LEVEL
B-Y PHASE
OUTPUT LEVEL

ABE #2 Board #5

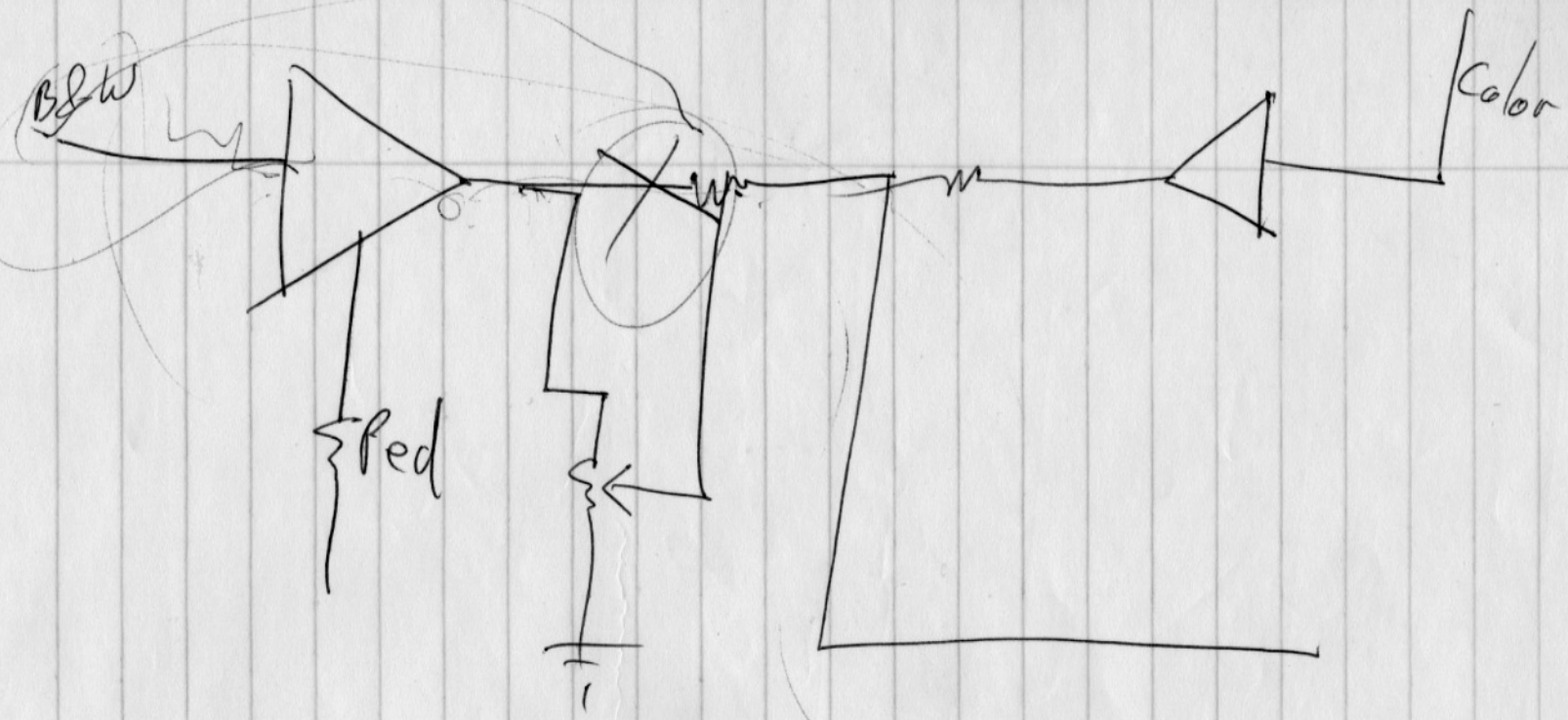


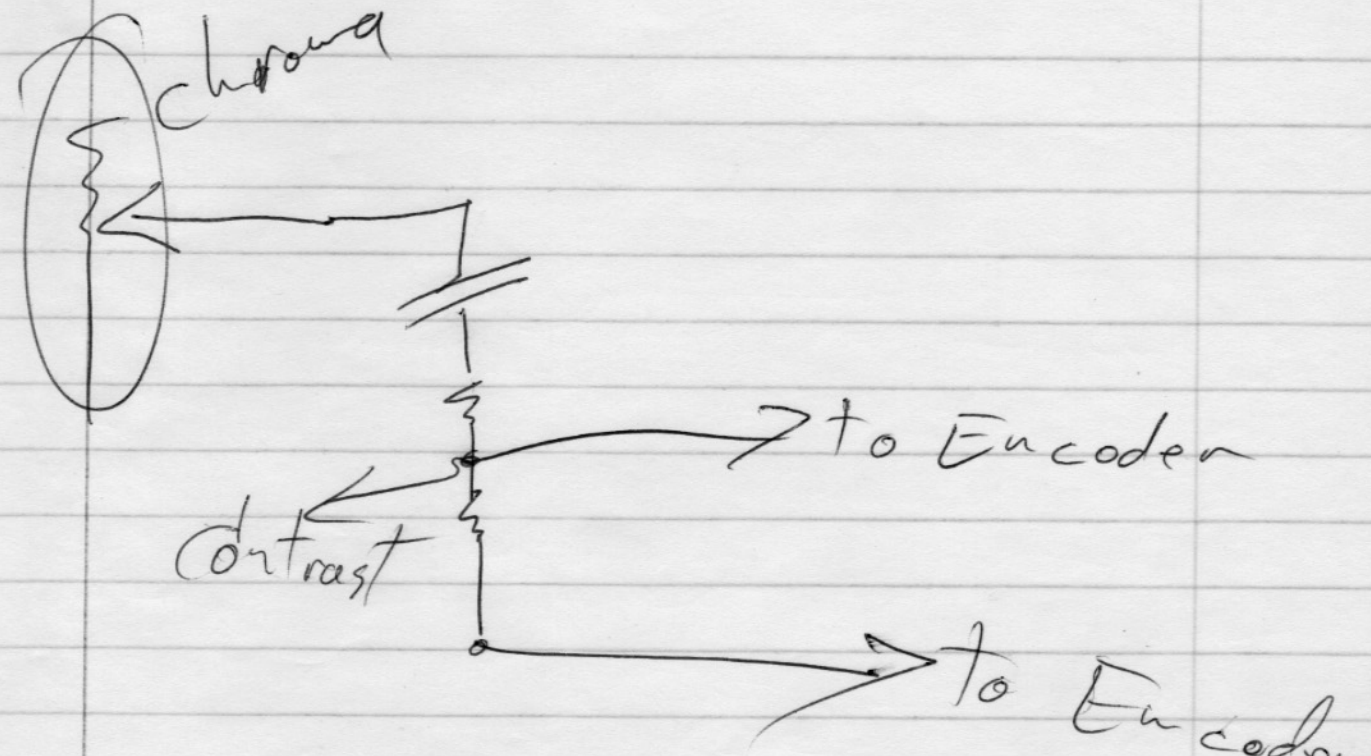
1 ⏚
2 -6
3 IN A
4 ⏚
5
6 out A
7 ⏚
8
9
10
11
12
13
14 ⏚
15
16 ⏚
17
18 ⏚
19 IN B
20 ⏚
21 +12
22 ⏚

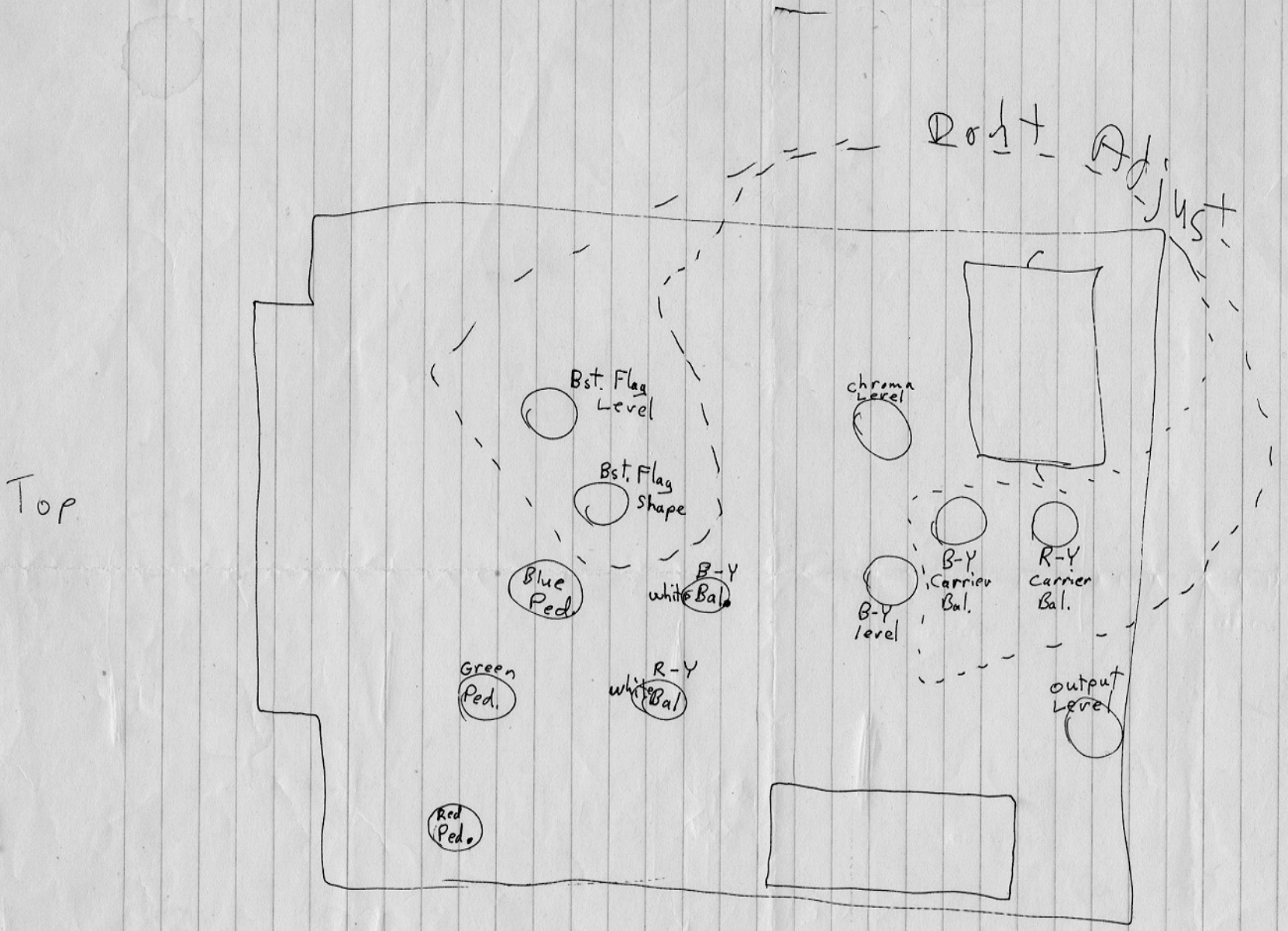
ABE #2 Board #3

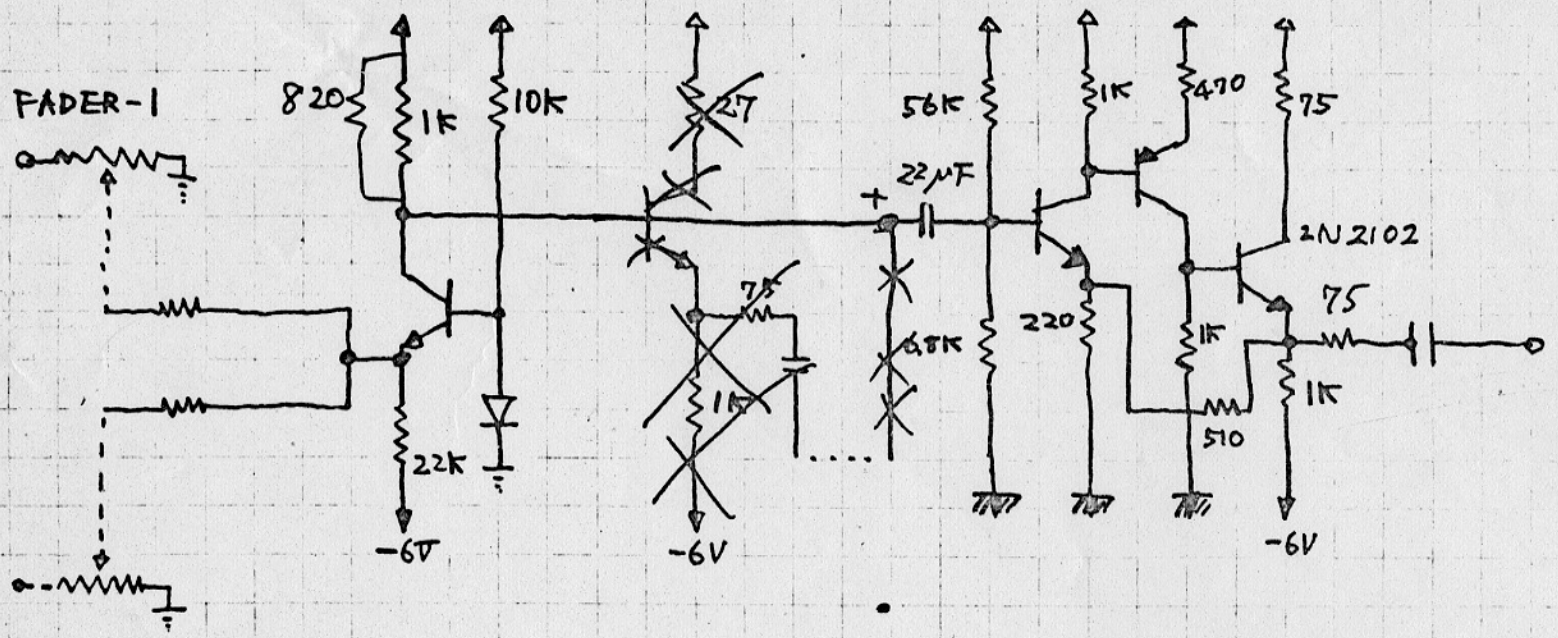

Pin3
220
82
470p
2.2K
82
100K
100K
2.2K
3.3K
56K
33µ

8.2K
2.2K
170
1000
.01
150
10K
47µf.


B&W
Color
Red


chroma
to Encoder
contrast
to Encoder


Top
Roll Adjust
Bst. Flag Level
Bst. Flag Shape
Blue Ped.
B-Y white Bal.
Green Ped.
R-Y white Bal.
Red Ped.
chroma Level
B-Y level
B-Y Carrier Bal.
R-Y carrier Bal.
output Level



FADER - 2

PNP ... 2N4402 or Equivalent

NPN ... 2N4400 " "

MIXER - FADER OUTPUT AMP